### 3.6.5 Solving Problems Involving Quadratic Equations

**Steps to solve a problem**

**Step 1:** Convert the word problem to a quadratic equation form

**Step 2:** Solve the quadratic equation obtained in any one of the above three methods.

**Step 3:** Relate the mathematical solution obtained to the statement asked in the question.

**Example 3.36** The product of Kumaran's age (in years) two years ago and his age four years from now is one more than twice his present age. What is his present age?

***Solution*** Let the present age of Kumaran be $x$ years.

Two years ago, his age $= (x - 2)$ years.

Four years from now, his age $= (x + 4)$ years.

Given, $(x-2)(x+4) = 1 + 2x$

$x^2 + 2x - 8 = 1 + 2x \Rightarrow (x-3)(x+3) = 0$ then, $x = \pm 3$

Therefore, $x = 3$ (Rejecting $-3$ as age cannot be negative)

Kumaran's present age is 3 years.

**Example 3.37** A ladder 17 feet long is leaning against a wall. If the ladder, vertical wall and the floor from the bottom of the wall to the ladder form a right triangle, find the height of the wall where the top of the ladder meets if the distance between bottom of the wall to bottom of the ladder is 7 feet less than the height of the wall?

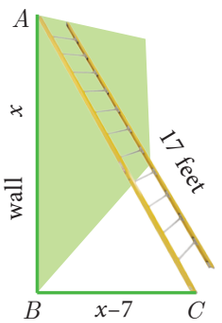


Fig. 3.7

***Solution*** Let the height of the wall $AB = x$ feet

As per the given data $BC = (x–7)$ feet

In the right triangle $ABC$, $AC = 17$ ft, $BC = (x–7)$ feet

By Pythagoras theorem, $AC^2 = AB^2 + BC^2$

$(17)^2 = x^2 + (x-7)^2$; $289 = x^2 + x^2 - 14x + 49$

$x^2 - 7x - 120 = 0$ hence, $(x-15)(x+8) = 0$ then, $x = 15$ (or) $-8$

Therefore, height of the wall AB $= 15$ ft (Rejecting $-8$ as height cannot be negative)

**Example 3.38** A flock of swans contained $x^2$ members. As the clouds gathered, $10x$ went to a lake and one-eighth of the members flew away to a garden. The remaining three pairs played about in the water. How many swans were there in total?

***Solution*** As given there are $x^2$ swans.

As per the given data $x^2 - 10x - \frac{1}{8}x^2 = 6$ we get, $7x^2 - 80x - 48 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{80 \pm \sqrt{6400 - 4(7)(-48)}}{14} = \frac{80 \pm 88}{14}$$



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